

Every ordinal as a point-separating game value

Lexicographic cubes, compact T_1 realizations,
and a proposed solution of a recent spectrum problem

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Abstract

For every ordinal λ , let $\mathbb{B}_\lambda = \{0, 1\}^\lambda$ carry its lexicographic order. We prove that its topology of all downward-closed sets has point-separating game value exactly λ . The lower bound has a particularly short strategy: after each query, retain an entire immediate successor cylinder. At limit stages, take the union of the retained prefixes. We then prove the same exact value for the topology generated by downward-closed sets with finitely many points deleted. This second topology is compact and T_1 . Its lower bound uses a finite-prefix homogeneity lemma and transfinite fusion in blocks of length ω . Consequently every ordinal occurs as the point-separating value of a compact T_1 space. In particular, the value $\omega^2 + 1$ gives an affirmative answer to Problem 1.2 of Chiozini, Csernák, and Soukup, in its stated generality. The infinite examples are not Hausdorff. We also give an order-embedding characterization of the invariant for downward topologies, exact nonadaptive separation numbers in the countable case, and explicit winning strategies for both players on the appropriate sides of the threshold.

Status and scope. This is a proposed research solution with full proofs, not an independently refereed or machine-formalized result. The checked source is arXiv:2510.05754v3. The construction answers its unrestricted topological-space question and even yields compact T_1 witnesses. It does *not* establish realization by Hausdorff, regular Hausdorff, or zero-dimensional Hausdorff spaces. Throughout this manuscript, compactness means the finite-subcover property; Hausdorffness is a separate condition.

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1 The selected problem and the main result

1.1 A concrete ordinal-spectrum question

Chiozini, Csernák, and Soukup introduce an ordinal-valued point-separating game invariant in [1]. Their Problem 1.2 asks whether a topological space can satisfy

$$\omega^2 < \text{ps}(X) < \omega_1. \quad (1)$$

The question is posed in a framework allowing T_0 spaces. The same source constructs zero-dimensional Hausdorff examples for all values at most ω^2 . The version checked here, v3, was submitted on 29 May 2026; its PDF is dated 1 June 2026 and still explicitly poses the question on pages 4 and 16. The bibliographic record retains the earlier title *Cut-and-choose games in topological spaces*, while the v3 PDF has the title recorded in our reference.

Our approach replaces complicated topological constructions by an ordered search space. A binary sequence indexed by an ordinal can be read one coordinate at a time. The point is to choose a topology in which no alternative query can uniformly finish earlier. Downward-closed queries have exactly the required property. Finite deletions permit a T_1 refinement without destroying the transfinite obstruction.

1.2 The construction

For an ordinal λ , define

$$\mathbb{B}_\lambda = \{x : x : \lambda \longrightarrow \{0, 1\}\}.$$

If $x \neq y$, let ξ be their first coordinate of disagreement. Set $x <_{\text{lex}} y$ when $x(\xi) = 0$ and $y(\xi) = 1$. This is a linear order because λ is well ordered.

On this ordered set, use the following two topologies:

$$\tau_\downarrow = \{D \subseteq \mathbb{B}_\lambda : y \leq_{\text{lex}} x \in D \implies y \in D\}, \quad (2)$$

$$\tau_{\text{fin}} = \left\{ \bigcup_{i \in I} (D_i \setminus F_i) : D_i \in \tau_\downarrow, F_i \subseteq \mathbb{B}_\lambda \text{ finite} \right\}. \quad (3)$$

Write $X_\lambda^\downarrow = (\mathbb{B}_\lambda, \tau_\downarrow)$ and $X_\lambda^{\text{fin}} = (\mathbb{B}_\lambda, \tau_{\text{fin}})$.

Theorem 1.1 (Ordinal realization). *For every ordinal λ ,*

$$\text{ps}(X_\lambda^\downarrow) = \text{ps}(X_\lambda^{\text{fin}}) = \lambda. \quad (4)$$

The space $X_\lambda^\downarrow$ is compact and T_0 , and X_λ^{fin} is compact and T_1 . For either space, Seeker has a winning strategy in the length- β point-separating game exactly when $\beta \geq \lambda$; Hider has a winning strategy exactly when $\beta < \lambda$. For infinite λ , neither construction is Hausdorff.

Corollary 1.2. *There is a compact T_1 space of cardinality $\mathfrak{c} = 2^{\aleph_0}$ satisfying $\text{ps}(X) = \omega^2 + 1$. More generally, every countably infinite ordinal occurs as $\text{ps}(X)$ for a compact T_1 space of cardinality $\mathfrak{c}$.*

Proof. Take $X = X_{\omega^2+1}^{\text{fin}}$, or $X = X_\lambda^{\text{fin}}$ for an arbitrary countably infinite λ . The underlying set has cardinality $2^{|\lambda|} = \mathfrak{c}$. Apply [theorem 1.1](#). $\square$

No continuum hypothesis, forcing assumption, or large-cardinal hypothesis is used. The theorem is formulated for all set ordinals, not only countable ones. Its primary claim of research interest is the realization statement in relation to (1); the elementary order-theoretic consequences below are not claimed to be historically new in isolation.

2 Game conventions and ordinal preliminaries

2.1 The point-separating game

We use the point-separating game of [1, Definition 1.1]. Fix a topological space X and an ordinal β . At turn $\xi < \beta$, Seeker chooses an open set U_ξ , and Hider answers $i_\xi \in \{0, 1\}$. Put

$$V_\xi = \begin{cases} U_\xi & i_\xi = 1, \\ X \setminus U_\xi & i_\xi = 0, \end{cases} \quad W_\eta = \bigcap_{\xi < \eta} V_\xi \quad (\eta \leq \beta),$$

with $W_0 = X$. Seeker wins if $|W_\beta| \leq 1$; otherwise Hider wins. The point-separating number is

$$\text{ps}(X) = \min\{\beta : \text{Seeker has a winning strategy in } G_\beta(X)\}.$$

A strategy may depend on the complete earlier history, including at limit stages. No computability restriction is imposed.

If the candidate set ever becomes empty, it remains empty, and Seeker wins. Thus a lower bound must preserve actual points through limit intersections. Showing only that every finite position is nonempty is insufficient. All lower bounds in this manuscript maintain an explicit full cylinder inside the candidate set, or a specified finite set at the very end.

There are two distinct logical assertions worth keeping separate:

$$\text{Seeker has no winning strategy,} \quad \text{Hider has a winning strategy.}$$

The second implies the first, but no general determinacy principle is being assumed. We prove the stronger assertion directly for our spaces. The strategies therefore also refute every fixed Seeker strategy in the interpretation where a hidden point is chosen in advance: the resulting final candidate set contains two points, either of which could have generated all the recorded answers.

2.2 Ordinal addition is not cardinal arithmetic

All additions and multiplications of ordinals below are the ordinary ordinal operations, not the natural operations. In particular, $\omega \cdot \alpha$ means α consecutive blocks of order type ω .

Every ordinal λ has a unique decomposition

$$\lambda = \omega \cdot \alpha + n, \quad n < \omega. \quad (5)$$

This is division by ω on the left. One way to obtain it is to take the largest α with $\omega \cdot \alpha \leq \lambda$. Such an α exists because multiplication by ω on the left is increasing and continuous in the right argument, and $\omega \cdot \alpha \geq \alpha$ bounds the relevant set of candidates. The remaining interval has order type less than ω . Uniqueness follows from $\omega \cdot \alpha + n < \omega \cdot (\alpha + 1)$.

At a limit ordinal δ ,

$$\sup_{\xi < \delta} \omega \cdot \xi = \omega \cdot \delta. \quad (6)$$

For a fixed ξ , a strictly increasing sequence of finite integers k_j satisfies

$$\sup_{j < \omega} (\omega \cdot \xi + k_j) = \omega \cdot (\xi + 1).$$

These two observations account for all limit-stage bookkeeping in the T_1 proof. Merely enumerating a countable ordinal by $\mathbb{N}$ would not preserve this order of dependence.

3 Lexicographic cylinders and the two topologies

3.1 Prefix geometry

For $\eta \leq \lambda$ and $s \in \{0, 1\}^\eta$, define the full cylinder

$$C_s = \{x \in \mathbb{B}_\lambda : x \upharpoonright \eta = s\}. \quad (7)$$

The length of s is its ordinal domain, denoted $\text{lh}(s)$. The empty prefix has cylinder $\mathbb{B}_\lambda$. A prefix of length λ has a singleton cylinder. Every shorter prefix has at least two extensions.

Lemma 3.1 (Cylinder geometry). *The cylinders have the following properties.*

- (a) *Each C_s is nonempty and convex in the lexicographic order.*
- (b) *If $\text{lh}(s) < \lambda$, then $C_s = C_{s \smallfrown 0} \dot{\cup} C_{s \smallfrown 1}$, and every element of the first child is less than every element of the second.*
- (c) *For an increasing compatible family of prefixes $(s_\xi)_{\xi < \delta}$, with union s , one has $\bigcap_{\xi < \delta} C_{s_\xi} = C_s$.*

Proof. A prefix extends to a point of $\mathbb{B}_\lambda$ by assigning zero to all remaining coordinates. If two extensions of the same prefix bracket a third point, that point cannot first disagree with the prefix before its end, proving convexity. The second statement follows by inspection of the next bit. For the last statement, belonging to every cylinder is exactly the condition of agreeing with every s_ξ , hence with their union. $\square$

Notice that these are *full* cylinders: no restrictions are placed on later coordinates. This is what prevents a hidden loss of choices at a limit.

3.2 The downward topology

The family $\tau_\downarrow$ is closed under arbitrary unions and arbitrary intersections, and contains the empty set and the whole space. It is therefore a topology, indeed an Alexandrov topology. This use of “Alexandrov” means closure under arbitrary intersections of open sets; it does not refer to the double-arrow space.

For $x <_{\text{lex}} y$, the principal initial segment $\{z : z \leq_{\text{lex}} x\}$ contains x but not y , so $X_\lambda^\downarrow$ is T_0 . Its greatest point is 1^λ . The only downward-closed open set containing this point is the whole space, so every open cover contains the whole space as one of its members. In particular, $X_\lambda^\downarrow$ is compact.

If $\lambda > 0$, every nonempty downward-open set contains 0^λ . Thus $X_\lambda^\downarrow$ is not T_1 and is not Hausdorff. This simple topology will first provide a transparent proof of the ordinal obstruction.

3.3 The finite-deletion refinement

The sets $D \setminus F$ in (3) form a base because

$$(D \setminus F) \cap (E \setminus G) = (D \cap E) \setminus (F \cup G).$$

This is the smallest topology containing the downward topology and all cofinite sets.

Proposition 3.2. *For every ordinal λ , X_λ^{fin} is compact and T_1 . It is discrete when λ is finite and is not Hausdorff when λ is infinite.*

Proof. For every x , the set $\mathbb{B}_\lambda \setminus \{x\}$ is basic open, so all singletons are closed. This proves T_1 .

Let an open cover be given, and choose a member U containing 1^λ . There is a basic set $D \setminus F$ with $1^\lambda \in D \setminus F \subseteq U$. Downward closure forces $D = \mathbb{B}_\lambda$, so U misses only finitely many points. At most one further cover member per missing point gives a finite subcover.

When λ is finite, the underlying set is finite and every subset is cofinite, so the topology is discrete. Now suppose $\lambda \geq \omega$. Let y have first bit 1 and all later bits 0, and let $t = 1^\lambda$. Then $y \neq t$, and infinitely many points lie below y . Every neighborhood of y contains all those predecessors with at most finitely many exceptions. Every neighborhood of t is cofinite, by the preceding argument. These two types of neighborhoods must intersect, so y and t have no disjoint neighborhoods. $\square$

Neither topology should be confused with the product topology on $\{0, 1\}^\lambda$ or with its two-sided lexicographic order topology. The exact collection of legal queries is essential to the result.

4 An exact value for downward queries

4.1 Seeker's coordinate strategy

For a prefix s of length $\eta < \lambda$, define

$$D_s = \{x : x \upharpoonright \eta <_{\text{lex}} s\} \cup \{x : x \upharpoonright \eta = s \text{ and } x(\eta) = 0\}. \quad (8)$$

This is a downward-closed subset of the *whole* ordered space, and

$$D_s \cap C_s = C_{s \smallfrown 0}. \quad (9)$$

Proposition 4.1 (Upper bound). *Seeker wins $G_\lambda(X_\lambda^\downarrow)$ and $G_\lambda(X_\lambda^{\text{fin}})$.*

Proof. Start with the empty prefix. At turn η , query D_s , where s is the prefix determined by the previous answers. An answer 1 determines the next bit to be 0, and an answer 0 determines it to be 1. Thus the candidate set is exactly the new cylinder. At a limit, take the union of the earlier prefixes; lemma 3.1(c) gives the same invariant. After λ turns the prefix has length λ , so exactly one point remains. Each query belongs to both topologies. $\square$

In particular, the strategy does not ask the globally defined question “is coordinate η zero?” without qualification. That set need not be open. Instead it uses the global initial segment (8), whose trace on the currently known cylinder is precisely the desired question.

4.2 Hider's one-bit strategy

Lemma 4.2 (An untouched child). *Let D be downward closed and let s be a prefix of length less than λ . Then either*

$$C_{s \smallfrown 0} \subseteq D \quad \text{or} \quad C_{s \smallfrown 1} \subseteq \mathbb{B}_\lambda \setminus D.$$

Proof. If the first inclusion fails, choose $a \in C_{s \smallfrown 0} \setminus D$. Every $b \in C_{s \smallfrown 1}$ is greater than a . If such a b belonged to D , downward closure would put a in D , a contradiction. $\square$

Theorem 4.3 (Downward-query threshold). *For $\beta < \lambda$, Hider has a winning strategy in $G_\beta(X_\lambda^\downarrow)$. Consequently $\text{ps}(X_\lambda^\downarrow) = \lambda$.*

Proof. Maintain a prefix s_η of length exactly η with

$$C_{s_\eta} \subseteq W_\eta. \quad (10)$$

Initially this holds for the empty prefix. When Seeker asks D at stage η , use the following rule:

condition	answer	retained prefix
$C_{s_\eta \frown 0} \subseteq D$	1	$s_\eta \frown 0$
$C_{s_\eta \frown 0} \not\subseteq D$	0	$s_\eta \frown 1$

The second case is legitimate by [lemma 4.2](#). In either case the new cylinder is contained in the old candidate set and in the chosen side of the query.

At a limit η , define $s_\eta = \bigcup_{\xi < \eta} s_\xi$. Its length is η , and [lemma 3.1\(c\)](#) proves (10); in particular the intersection is not empty. At the end, C_{s_β} still has a free coordinate because $\beta < \lambda$. It contains two distinct points, so Hider wins. The upper bound is [proposition 4.1](#). $\square$

This proof is entirely online: Hider uses only the current query and the retained prefix. It does not inspect a future decision tree or invoke a determinacy theorem. It already proves [corollary 1.2](#) with “compact T_0 ” in place of “compact T_1 ”.

5 Preserving the value under the T_1 refinement

A finite deletion can destroy the untouched-child argument. For example, a query might delete one point from each immediate child. Thus the preceding proof cannot simply be reused after refining the topology. The replacement allows finitely many prefix bits to be spent on a single move, while ensuring that only one block of ω bits is spent over ω moves.

5.1 Finite-prefix homogeneity

Lemma 5.1 (Homogeneous finite extension). *Let s have length θ , where $\theta + \omega \leq \lambda$, and let $U \in \tau_{\text{fin}}$. There is a nonempty finite binary word v such that*

$$C_{s \frown v} \subseteq U \quad \text{or} \quad C_{s \frown v} \subseteq \mathbb{B}_\lambda \setminus U. \quad (11)$$

Proof. First suppose every point of $U \cap C_s$ has zero in all of the next ω coordinates. Then $C_{s \frown 1}$ is disjoint from U , and we are done.

Otherwise choose $x \in U \cap C_s$ with a 1 among the next ω bits. Let $j < \omega$ be the first such position. Thus the next j bits are zero and $x(\theta + j) = 1$. Choose a basic neighborhood

$$x \in D \setminus F \subseteq U,$$

where D is downward closed and F is finite. Put $p = s \frown 0^{j+1}$. Every point of C_p is strictly below x , so $C_p \subseteq D$.

Choose a finite k with $2^k > |F|$. The 2^k cylinders

$$C_{p \frown w} \quad (w \in \{0, 1\}^k)$$

are pairwise disjoint and nonempty. At most $|F|$ of them meet F . Choose one that does not, say $C_{p \frown w}$. It lies in $D \setminus F \subseteq U$. The word $v = 0^{j+1} \frown w$ is finite and nonempty. All coordinates used exist because $\theta + j + 1 + k < \theta + \omega \leq \lambda$. $\square$

Remark 5.2 (Making the rule a strategy). Among all nonempty finite words satisfying (11), select one of shortest length and then the lexicographically first of that length. This is a definite choice rule depending only on s and U . Answer according to the side containing its cylinder. The lemma guarantees that the rule is always defined when a full ω -block remains. Testing the inclusion of an infinite cylinder in an arbitrary open set is not claimed to be an effective computation; strategies here are set-theoretic functions, as in the original game.

5.2 Fusion through one block and through limit blocks

Suppose a full cylinder with prefix length $\omega \cdot \xi$ is contained in the candidate set at game time $\omega \cdot \xi$, and $\omega \cdot (\xi + 1) \leq \lambda$. Apply lemma 5.1 at each of the next finitely indexed turns. After $r < \omega$ turns of this block, the tracked prefix has length

$$\omega \cdot \xi + k_r, \quad k_r < \omega, \quad k_0 = 0, \quad k_{r+1} > k_r. \quad (12)$$

The applicability condition remains true: adding ω to any length in (12) gives $\omega \cdot (\xi + 1)$.

At the end of the block, the union of the prefixes has length exactly $\omega \cdot (\xi + 1)$, because the strictly increasing finite sequence (k_r) is unbounded in ω . Its full cylinder survives. At a limit δ of blocks, take the union of all earlier tracked prefixes. Equation (6) shows that the resulting length is $\omega \cdot \delta$, and lemma 3.1(c) again shows that its full cylinder survives.

The construction therefore satisfies two different invariants. At block boundaries, game time and prefix length agree. Within a block, prefix length may run finitely far ahead of game time. It can never reach the end of that ω -block after only finitely many moves.

5.3 The finite terminal block

Lemma 5.3 (Finite survival). *If a candidate set contains a specified set A of 2^n points, Hider can retain at least 2^{n-r} of these points after any $r \leq n$ further binary queries, regardless of which subsets are allowed as queries.*

Proof. For each query retain the larger of its two intersections with the current tracked finite set. A set of at least 2^m points has one side with at least 2^{m-1} points. Induction proves the assertion. $\square$

Theorem 5.4 (Finite-deletion threshold). *For $\beta < \lambda$, Hider has a winning strategy in $G_\beta(X_\lambda^{\text{fin}})$. Hence $\text{ps}(X_\lambda^{\text{fin}}) = \lambda$.*

Proof. Write $\lambda = \omega \cdot \alpha + n$ with $n < \omega$. During the first $\omega \cdot \alpha$ moves, use the homogeneous-extension strategy and the fusion rule just described. At time $\omega \cdot \xi$ for $\xi \leq \alpha$, it preserves a full cylinder with prefix length $\omega \cdot \xi$.

Consider any shorter stopping time $\beta = \omega \cdot \delta + r < \lambda$, where $r < \omega$. Necessarily $\delta \leq \alpha$. If $\delta < \alpha$, only finitely many moves have been made in block δ . The tracked prefix has length $\omega \cdot \delta + k$ for some finite k . At least the rest of that infinite block is free, so its cylinder contains infinitely many points.

If $\delta = \alpha$, then $r < n$. At time $\omega \cdot \alpha$, the surviving full cylinder has exactly 2^n points, indexed by the remaining n bits. Apply lemma 5.3. After r more moves at least $2^{n-r} \geq 2$ points remain. This also handles $\alpha = 0$, when the whole space is finite.

Thus the specified Hider strategy wins at every $\beta < \lambda$. Proposition 4.1 supplies the matching Seeker strategy at λ . $\square$

Completion of the proof of theorem 1.1. The equalities follow from theorems 4.3 and 5.4. The separation and compactness properties were established in proposition 3.2 and the preceding subsection on

downward topologies. Seeker's strategy at λ extends to any longer game by making arbitrary additional queries: the candidate set cannot grow. The given Hider strategies apply at every shorter length. Both players cannot have winning strategies in the same game, since playing them against each other produces a single play with only one winner. $\square$

6 The first example beyond ω^2 , in full detail

Take

$$X = \left(\{0, 1\}^{\omega^2+1}, \tau_{\text{fin}} \right). \quad (13)$$

The coordinates before ω^2 are arranged as

$$[0, \omega), [\omega, \omega \cdot 2), [\omega \cdot 2, \omega \cdot 3), \dots,$$

followed by the single coordinate ω^2 . This arrangement is an ordinal sequence of blocks, not an arbitrary enumeration of a countable set.

Seeker can read the first block in ω moves, the second in the next ω moves, and so on, using (8). At time ω^2 , all coordinates below ω^2 have been determined. One further initial-segment query determines the last bit. This gives $\text{ps}(X) \leq \omega^2 + 1$.

Against any Seeker play of length ω^2 , Hider applies the finite-prefix homogeneity lemma during the first block and retains a full cylinder with an ω -long prefix. Repeating the procedure, at time $\omega \cdot m$ a full cylinder with prefix length $\omega \cdot m$ survives. At time ω^2 , the union prefix s has length ω^2 and

$$\{s \frown 0, s \frown 1\} \subseteq W_{\omega^2}.$$

The last bit has not been spent. Therefore Hider wins $G_{\omega^2}(X)$. The same strategy also wins at every earlier stopping time, so

$$\text{ps}(X) = \omega^2 + 1.$$

This witness is compact and T_1 by [proposition 3.2](#), has size $\mathfrak{c}$, and is not Hausdorff. In the downward topology, an even simpler account is available: Hider retains a cylinder with prefix length exactly equal to game time, so the two final extensions survive directly by [theorem 4.3](#).

Target value λ	Underlying ordered set	ω -block decomposition
$n < \omega$	$\{0, 1\}^n$	$\omega \cdot 0 + n$
$\omega + 3$	$\{0, 1\}^{\omega+3}$	$\omega \cdot 1 + 3$
ω^2	$\{0, 1\}^{\omega^2}$	$\omega \cdot \omega$
$\omega^2 + 1$	$\{0, 1\}^{\omega^2+1}$	$\omega \cdot \omega + 1$
$\omega^3 + 7$	$\{0, 1\}^{\omega^3+7}$	$\omega \cdot \omega^2 + 7$
ω^ω	$\{0, 1\}^{\omega^\omega}$	$\omega \cdot \omega^\omega$

Table 1: For every row, either of the two specified topologies has value λ ; the finite-deletion topology is compact and T_1 . The last equality uses $1 + \omega = \omega$ in the exponent of an ordinal power.

7 An order-theoretic characterization

The same game describes exactly the least length of a lexicographic binary code for an arbitrary linear order. This gives a direct connection with order types, rather than only with topological invariants.

For a linear order L , let $L^\downarrow$ denote L with all downward-closed sets open. Its game value exists: well-order the set of principal initial segments and ask all of them. These sets separate every pair of distinct points.

Theorem 7.1 (Lexicographic coding). *For every linear order L and every ordinal β , the following are equivalent:*

- (i) *Seeker has a winning strategy in $G_\beta(L^\downarrow)$.*
- (ii) *There is an order embedding $L \hookrightarrow (\mathbb{B}_\beta, <_{\text{lex}})$.*

Consequently

$$\text{ps}(L^\downarrow) = \min\{\beta : L \text{ order-embeds into } \mathbb{B}_\beta\}. \quad (14)$$

Proof. Suppose first that a winning strategy is given. For each $x \in L$, follow the play in which Hider always chooses the side containing x . Encode the answer “inside” by 0 and “outside” by 1, obtaining a word $c(x) \in \{0, 1\}^\beta$. Distinct points have distinct words, since otherwise both would survive the same complete play.

Let $x < y$. At their first different answer, the preceding histories agree, so the strategy asks the same downward-closed set D . It is impossible that $x \notin D$ and $y \in D$. Therefore x answers inside and y answers outside. In our coding convention the first different bits are 0 and 1, respectively. Hence $c(x) <_{\text{lex}} c(y)$, and c is an order embedding.

Conversely, let $e : L \hookrightarrow \mathbb{B}_\beta$ be an order embedding. For a current code prefix s of length $\eta < \beta$, ask $e^{-1}(D_s)$, where D_s is the initial segment (8) in $\mathbb{B}_\beta$. This preimage is downward closed in L . The response determines one more code bit. After β moves, all remaining points have the same code, and injectivity of e leaves at most one. Impossible code histories simply give an empty candidate set, which also wins for Seeker. $\square$

Corollary 7.2 (Strict hierarchy of order types). *For any ordinals α, β ,*

$$\mathbb{B}_\alpha \text{ order-embeds into } \mathbb{B}_\beta \iff \alpha \leq \beta.$$

Proof. If $\alpha \leq \beta$, extend each α -sequence by zeros to length β ; its first disagreement with another sequence is unchanged. Conversely, an embedding with $\beta < \alpha$ would give a winning length- β strategy on $X_\alpha^\downarrow$ by [theorem 7.1](#), contradicting [theorem 4.3](#). $\square$

Thus the countably infinite lexicographic cubes form a strictly increasing order-embedding hierarchy, even though all have the same cardinality $\mathfrak{c}$. Their game values measure this order-sensitive coding length. A bijection between the coordinate sets of two countable ordinals does not in general preserve their lexicographic order.

8 Adaptive versus nonadaptive separation

For a T_0 space X , define

$$\text{sep}(X) = \min\{|\mathcal{U}| : \mathcal{U} \subseteq \tau_X, \forall x \neq y \exists U \in \mathcal{U} (|U \cap \{x, y\}| = 1)\}.$$

Unlike ps , this is a cardinal: all queries in $\mathcal{U}$ are chosen in advance. For infinite spaces, this agrees with the T_0 -pseudoweight discussed in [1]; the definition above avoids finite-cardinal convention issues. Our calculations here are proved directly.

8.1 Adjacent pairs force one cut each

Let

$$\{0, 1\}^{<\lambda} = \bigcup_{\eta < \lambda} \{0, 1\}^\eta$$

be the set of all proper-length prefixes, with sequences of different lengths regarded as different elements.

Proposition 8.1. *For every ordinal λ ,*

$$\text{sep}(X_\lambda^\perp) = |\{0, 1\}^{<\lambda}|. \quad (15)$$

In particular this equals $2^n - 1$ for $\lambda = n < \omega$, equals $\aleph_0$ for $\lambda = \omega$, and equals $\mathfrak{c}$ for $\omega < \lambda < \omega_1$.

Proof. The family $(D_s)_{s \in \{0,1\}^{<\lambda}}$ separates all pairs: use their first coordinate of disagreement. This gives the upper bound.

For each proper prefix s of length η , form the pair

$$a_s = s \frown 0 \frown 1^{[\eta+1, \lambda)}, \quad b_s = s \frown 1 \frown 0^{[\eta+1, \lambda)}.$$

Here the last notation means assigning the indicated constant value to all remaining coordinates. The point a_s is the maximum of $C_{s \frown 0}$, and b_s is the minimum of $C_{s \frown 1}$. They are adjacent: no point lies strictly between them. Different prefixes give different adjacent pairs.

A downward-closed set can separate at most one adjacent pair in a linear order. Indeed, if the first pair occurs before the second, containing the lower point of the second pair forces containment of the upper point of the first pair. Every separating family must separate all these pairs, proving the lower bound. The stated special cases follow by counting the levels of the prefix tree. $\square$

8.2 Finite deletions can split only countably many ordered fibres

The following lemma handles arbitrary unions of finite-deletion basic sets; it does not incorrectly assume that an arbitrary open set itself has only finitely many holes.

Lemma 8.2 (Split fibres). *Let L be a linear order with the topology generated by downward-closed sets minus finite sets. Suppose $(A_t)_{t \in I}$ is a family of nonempty subsets indexed by a linear order, with every point of A_s below every point of A_t whenever $s < t$. For an open set U , let*

$$E = \{t : \emptyset \neq U \cap A_t \neq A_t\}.$$

Then E is at most countable. More precisely, every element of E has only finitely many predecessors in E .

Proof. Fix $t \in E$, choose $b \in U \cap A_t$, and take $b \in D \setminus F \subseteq U$ with F finite. For every earlier $s \in E$, there is a point $a_s \in A_s \setminus U$. Such a point lies below b , hence in D , and therefore in F . Distinct fibres supply distinct points, so there are at most $|F|$ such s .

In any linear order whose strict initial segments are finite, the map $t \mapsto |\{s \in E : s < t\}|$ is strictly increasing into $\mathbb{N}$. It is injective, so E is countable. $\square$

Proposition 8.3. *For countable ordinals λ ,*

$$\text{sep}(X_\lambda^{\text{fin}}) = \begin{cases} n, & \lambda = n < \omega, \\ \aleph_0, & \lambda = \omega, \\ \mathfrak{c}, & \omega < \lambda < \omega_1. \end{cases}$$

Proof. For finite n , the space is discrete with 2^n points. The n bit questions separate it, and fewer binary questions produce fewer than 2^n answer words. At $\lambda = \omega$, the countable family from [proposition 8.1](#) remains available, whereas no finite family separates an infinite set.

Now suppose $\omega < \lambda < \omega_1$. Partition $\mathbb{B}_\lambda$ into the ordered fibres $A_u = C_u$, $u \in \{0, 1\}^\omega$. Each fibre contains at least two points. For every u , a separating family must contain a set that splits A_u . By [lemma 8.2](#), one open set splits at most countably many fibres. There are $\mathfrak{c}$ fibres. Thus a separating family has cardinality at least $\mathfrak{c}$, since the union of κ countable sets has size at most $\max(\kappa, \aleph_0)$ for infinite κ . The family from [proposition 8.1](#) gives the matching upper bound. $\square$

Consequently, among the countable values strictly above ω , both constructions keep cardinality and nonadaptive separation number fixed at $\mathfrak{c}$, while ps takes every such ordinal. The additional information is not a cardinal-size effect; it records which queries can be made only after an earlier transfinite prefix has become known.

9 Verification, limitations, and further questions

9.1 What the accompanying computation checks

The proofs above do not rely on an empirical extrapolation. The accompanying Python program checks finite statements used to audit the reasoning:

- (a) the optimal comparison-query recurrence for chains of sizes up to 512;
- (b) the untouched-child rule for every prefix and every initial-segment query in all binary cubes of lengths 1 through 10;
- (c) the exact finite nonadaptive counts and the majority-side lower bound;
- (d) the finite pigeonhole assertion that more pairwise disjoint subcylinders than deleted points leave at least one subcylinder untouched.

The tests are exact integer and set computations, not floating-point experiments. The saved output records the bounds and counts. They do not verify transfinite induction, arbitrary open-set semantics, or the ZFC theorem in a proof assistant. Those obligations are addressed by the written proofs, especially [lemmas 3.1 and 5.1](#) and [theorem 5.4](#).

Finite grids must be interpreted carefully. Replacing an infinite binary block by a finite one makes the finite-deletion topology discrete on the whole finite space. Such a model is useful for the local pigeonhole argument but cannot be used as evidence that ordinal addition behaves like addition of finite query counts. In particular, the limit-stage proof cannot be inferred from numerical experiments.

9.2 The crucial proof obligations

The most important points for an independent check are the following.

Global legality of the queries. The set D_s in (8) is globally downward closed. A local coordinate condition on a known fibre is not silently assumed to be globally open.

Nonempty limits. Every retained prefix has all possible extensions. Increasing prefixes unite to a function on the union of their domains, and this function extends to the whole ordinal. Thus the intersection of the corresponding cylinders remains a full, nonempty cylinder.

Arbitrary refined opens. The finite-prefix lemma uses a basic neighborhood of just one point of an arbitrary open set. It therefore handles arbitrary unions of basic sets, not merely one downward set with one finite deletion.

Correct time accounting. In the refined topology, a single query may cost several prefix bits. Every cost is finite within the active ω -block. A strictly increasing sequence of finite costs fills exactly that block at its limit, and cannot spend any coordinate of the next block prematurely.

A genuine Hider strategy. The shortest-then-lexicographically-first homogeneous extension provides a rule for each current position and query. No future Seeker strategy is consulted. The last finite block uses a majority rule on tracked points. Thus the argument proves Hider wins, not merely the failure of one proposed search algorithm.

9.3 What is and is not settled

Within the stated T_0 framework, [theorem 1.1](#) gives an affirmative answer to (1), with the stronger compact T_1 conclusion. It does not settle a version requiring Hausdorff witnesses. In fact, [proposition 3.2](#) supplies an explicit obstruction to Hausdorffness for these very spaces.

There is a concrete reason that the proof cannot simply be transferred to an order topology. Two-sided open sets can cut independently inside many ordered fibres. For example, the disjoint union of three nonempty open intervals in an ordinary real line can split three disjoint, ordered intervals. This is incompatible with the finite-predecessor conclusion of [lemma 8.2](#). Thus replacing the topology changes the query model, rather than merely improving a separation axiom for free.

A natural next problem is to construct a Hausdorff space with value $\omega^2 + 1$, or to find a different invariant that obstructs this. A successful construction would need a replacement for the homogeneous-prefix lemma that survives the additional open sets required for Hausdorff separation. The present result identifies exactly what the one-sided ordered construction achieves, and exactly which extra requirement is not addressed.

The source check is limited: the identified v3 explicitly lists the spectrum question, and no later resolution was located in the searches performed for this manuscript. This does not certify priority against all publications or unpublished work. The proof is offered for mathematical checking on its own terms, with all substantive steps supplied.

10 Conclusion

The ordinal-indexed binary cube, equipped with downward-closed queries, has exact search length equal to its coordinate ordinal. The lower bound is not a cardinality argument: it comes from retaining a full successor cylinder at each turn and a full union cylinder at every limit. Passing to the cofinite refinement preserves the exact value by finite-prefix homogeneity and ω -block fusion, while making the space T_1 and retaining compactness.

The resulting statement is

Every ordinal is the point-separating value of a compact T_1 space.

In particular $X_{\omega^2+1}^{\text{fin}}$ is the explicit witness requested by the selected spectrum problem. The order-embedding characterization explains why these constructions naturally belong to the study of order types and transfinite notation: the game measures the least ordinal length of a lexicographic binary code.

A Strategy reference sheet

Seeker, both topologies, game length λ

At time η , retain the prefix determined by the previous answers, of length η . Ask D_s from (8). Append 0 after answer 1, and append 1 after answer 0. Unite the prefixes at limit stages. At time λ , the cylinder is a singleton.

Hider, downward topology, any game length below λ

Track a prefix s . If its left child is entirely contained in the query, keep that child and answer 1. Otherwise keep the right child and answer 0. Unite prefixes at limits. The prefix length equals game time, leaving a free coordinate at every shorter stopping time.

Hider, finite-deletion topology, any game length below λ

Write $\lambda = \omega \cdot \alpha + n$. Before time $\omega \cdot \alpha$, track a prefix inside the active infinite block and choose a shortest nonempty finite extension whose cylinder is homogeneous for the query. Such an extension exists by lemma 5.1. Answer for its containing side. Take unions at all limits. At time $\omega \cdot \alpha$, track all 2^n remaining points. For the last finite block choose the larger side at each query, breaking ties in a fixed way. Every stopping time before λ leaves at least two tracked points.

B Reproducing the archive

The archive contains this source, the compiled PDF, a BibTeX record of the primary reference, the standard-library Python verification program, its saved JSON output, and a separate proof-audit note. From the archive directory:

```
python3 code/finite_checks.py --output verification/results.json
pdflatex -interaction=nonstopmode -halt-on-error ordinal_separation.tex
pdflatex -interaction=nonstopmode -halt-on-error ordinal_separation.tex
```

The PDF uses an embedded bibliography, so a BibTeX run is unnecessary. A separate bibliography file, `references.bib`, is supplied for reuse. No third-party source PDF or font file is redistributed.

References

- [1] Lucas Chiozini, Tamás Csernák, and Lajos Soukup, *Gamification of the T_0 -pseudoweight via cut-and-choose games on topological spaces*, arXiv:2510.05754v3, submitted 29 May 2026, PDF dated 1 June 2026. The arXiv record uses the earlier title *Cut-and-choose games in topological spaces*. <https://arxiv.org/abs/2510.05754v3>; versioned PDF: <https://arxiv.org/pdf/2510.05754v3>. Definition 1.1, Problem 1.2, and the discussion of the ordinal spectrum. Accessed 20 September 2026.